

Dynamo on Intel XPU

Whitepaper

Dynamo on Intel XPU — Deployment Whitepaper

Why this document exists / scope note

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Dynamo on Intel XPU — Deployment Whitepaper

Scope: This document covers deploying Dynamo on Intel XPU (GPU) hardware in Kubernetes, using the official Intel XPU templates shipped in the Dynamo repository. It replaces informal/outdated internal notes on Dynamo + Intel XPU with a single source of truth, driven directly from the current state of the dynamo repository.

All content below is derived from reading the actual repository files (YAML templates, README.md files, docs/kubernetes/*). Nothing has been executed against a live Intel XPU Kubernetes cluster yet — see the Status column in each chapter and in tracking.md. Validate on real hardware before treating any command as final.

API version note: Chapter 1's 8 templates use the `nvidia.com/v1alpha1` CRD schema (`services: map`) — this is what the repo ships for all 8 Intel XPU patterns today. Dynamo is mid-migration to a newer `nvidia.com/v1beta1` schema (`components: list`, a real structural change, not just a version bump), but **XPU coverage under v1beta1 is only partial**: just 2 of the 8 patterns (agg, disagg) have been ported so far — see `to_fi.md` for the full

breakdown. If you're on a newer Dynamo Platform release that expects `v1beta1`, check `deploy/v1beta1/xpu/` for the latest state before assuming all 8 patterns are available there.

Why this document exists / scope note

Unlike `llm-d`, where most “well-lit path” guides ship their own dedicated `modelserver/xpu/...` overlay, **Dynamo's Intel XPU support is concentrated in a single directory**: `examples/backends/vllm/deploy/xpu/`. This directory ships **8 flat DRA (Dynamic Resource Allocation) YAML templates** covering combinations of serving pattern (aggregated / disaggregated) × optional add-ons (tracing, KV-aware routing, planner-based autoscaling). There is also one Intel XPU example under `examples/global_planner/`.

Repo-wide scan performed for this document (`grep -r xpu examples/docs/ recipes/` across the dynamo repository) found **no Intel XPU-specific content** under most other vLLM-backend templates, the SGLang backend, the TensorRT-LLM backend, and most of `recipes/`. It **did** find one Intel XPU recipe under `recipes/qwen3-vl-32b-fp8/` — a heterogeneous Intel XPU (encode) + NVIDIA GPU (decode) disaggregation, written up in full in Chapter 3. Chapter 4 distinguishes templates that look **feasible but simply haven't been templated for Intel XPU yet** (LoRA, GAIE, multi-node disaggregation, GMS/failover — all with no hard hardware blocker found) from the one **genuine NVIDIA-only dependency** found (TensorRT-LLM's proprietary compiler).

Bottom line: as of this writing, if you want to run Dynamo on Intel XPU, your options are the 8 templates in Chapter 1, the Global Planner example in Chapter 2, and the heterogeneous Qwen3-VL recipe in Chapter 3. Chapter 4 catalogs real candidates for future Intel XPU work (LoRA, GAIE, multi-node disagg, GMS/failover, SGLang) versus the one genuine NVIDIA-only dead end confirmed so far (TensorRT-LLM).

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Status icons: `verified working /` `documented from official`
 `sources but not yet run hands-on /` `not applicable (no Intel`
 `XPU support exists upstream)`

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Chapter 0 — Common Installation

All 8 templates in Chapter 1, plus the Global Planner example in Chapter 2, share the same cluster-level prerequisites. Do this section once per cluster.

0.1 Cluster Requirements

- **Kubernetes v1.34+** with the DRA (Dynamic Resource Allocation) `resource.k8s.io/v1` API enabled (DRA graduated to GA/stable behavior tracked by the templates; older DRA API versions such as `v1alpha3/v1beta1` are not compatible with the `apiVersion: resource.k8s.io/v1` used by every current template).
- Intel XPU (GPU) nodes, with the Intel GPU kernel driver installed on each node.
- **Intel resource drivers for Kubernetes** installed, exposing a DeviceClass named `gpu.intel.com`. Verify with:

```
kubectl get deviceclass gpu.intel.com
```

- Sufficient node resources for the model under test — the default model across all Chapter 1/2 templates is the small Qwen/Qwen3-0.6B, chosen deliberately by the upstream repo to keep XPU examples runnable on a single GPU.

0.2 Install the Dynamo Kubernetes Platform

Follow `docs/kubernetes/installation-guide.md` in the dynamo repo. Summary of the steps that apply regardless of accelerator vendor:

```
export NAMESPACE=dynamo-cloud
export RELEASE_VERSION=<dynamo-release-version> # e.g. v0.x.y,
see repo releases

# 1. Install core CRDs (cluster-scoped, install once per cluster)
helm install dynamo-crds \
  oci://ghcr.io/ai-dynamo/dynamo-crds \
  --version ${RELEASE_VERSION} \
  --namespace default

# 2. Install the Dynamo Platform (operator + supporting services)
into your namespace
kubectl create namespace ${NAMESPACE} 2>/dev/null || true
helm install dynamo-platform \
  oci://ghcr.io/ai-dynamo/dynamo-platform \
  --version ${RELEASE_VERSION} \
  --namespace ${NAMESPACE}
```

Intel XPU note: unlike NVIDIA deployments, you do **not** need the NVIDIA GPU Operator. Skip any GPU-Operator-specific step in the installation guide; the Intel device plugin/DRA driver from step 0.1 takes its place.

Confirm the operator is healthy:

```
kubectl get pods -n ${NAMESPACE}
kubectl get crd | grep dynamo
```

0.3 Build an Intel XPU vLLM Runtime Image

None of the 8 templates work with the stock `vllm-runtime` image — worker containers require a dedicated `vllm-runtime-xpu` image built with `VLLM_TARGET_DEVICE=xpu`. Build it from the repo's container/tooling:

```
cd dynamo/container
python render.py --framework=vllm --device=xpu --target=runtime
docker build -t <your-registry>/vllm-runtime-xpu:my-tag \
  -f vllm-runtime-xpu-amd64-rendered.Dockerfile .
docker push <your-registry>/vllm-runtime-xpu:my-tag
```

See `container/README.md` for the full, up-to-date build instructions and base image options. Substitute `<your-registry>/vllm-runtime-xpu:my-tag` for every occurrence of `nvcr.io/nvidia/ai-dynamo/vllm-runtime-xpu:my-tag` in the templates below, either by editing the YAML or by using `envsubst` with an image-name environment variable if your fork of the template parameterizes it.

The Frontend and Planner components do **not** need the XPU image — they run on the default `vllm-runtime` / `dynamo-planner` images (CPU-only control-plane components).

0.4 HuggingFace Token Secret

Every template references `envFromSecret: hf-token-secret`. Create it once per namespace:

```
export HF_TOKEN=<your-huggingface-token>
kubectl create secret generic hf-token-secret \
  --from-literal=HF_TOKEN=${HF_TOKEN} \
  -n ${NAMESPACE}
```

0.5 Optional: securityContext for Render/Video Group Access

Every worker container block in every template includes a commented-out securityContext snippet:

```
# NOTE: Uncomment if your environment requires specific group
access
# securityContext:
#   runAsUser: 1000
#   runAsGroup: 1000
#   supplementalGroups:
#     - 44 # render group
#     - 991 # video group
```

Uncomment and adjust the GIDs if your cluster's Pod Security Admission policy or your node image's /dev/dri device permissions require the container process to belong to specific render/video groups to access the Intel GPU. GIDs 44/991 are the upstream template's defaults and may differ from your actual node OS image — check `getent group render video` on a node.

1.1 Aggregated (agg_xpu_dra.yaml)

Status: documented from repo, not yet run hands-on.

Overview: The simplest pattern — a single `VllmDecodeWorker` role handles both prefill and decode for every request (no P/D separation). Good starting point to validate that DRA GPU allocation and the XPU runtime image work at all, before adding routing/tracing/disaggregation on top.

Prerequisites: Chapter 0 complete.

Deployment Steps:

```
export NAMESPACE=my-ns
kubectl apply -f examples/backends/vllm/deploy/xpu/
agg_xpu_dra.yaml -n ${NAMESPACE}
```

Verification:

```
kubectl get resourceclaim -n ${NAMESPACE}
kubectl get resourceslices
```

```
kubectl get dynamographdeployment -n ${NAMESPACE}
kubectl get pods -n ${NAMESPACE}
```

Inference Test:

```
kubectl port-forward deployment/vllm-agg-xpu-dra-frontend
8000:8000 -n ${NAMESPACE}
curl localhost:8000/v1/models
curl localhost:8000/v1/completions \
  -H "Content-Type: application/json" \
  -d '{"model": "Qwen/
Qwen3-0.6B", "prompt": "Hello", "max_tokens": 20}'
```

Troubleshooting: - If the VllmDecodeWorker pod stays Pending, check `kubectl describe resourceclaim` — a common cause is the Intel resource driver not yet reporting available devices (`kubectl get resourceslices` should list your XPU nodes). - If the worker crashes on startup complaining about missing device permissions, see §0.5.

Cleanup:

```
kubectl delete -f examples/backends/vllm/deploy/xpu/
agg_xpu_dra.yaml -n ${NAMESPACE}
```

Config Reference:

Field	Value	Notes
DynamoGraphDeployment name	vllm-agg-xpu-dra	
Services	Frontend (1 replica), VllmDecodeWorker (1 replica, aggregated)	
Model	Qwen/Qwen3-0.6B	override via <code>--model</code> arg
VLLM_TARGET_DEVICE	xpu	
GPU claim	1× <code>gpu.intel.com</code> per worker replica	
ephemeral-storage request	2Gi	increase for larger models

1.2 Aggregated + Tracing (agg_tracing_xpu_dra.yaml)

Status: documented from repo, not yet run hands-on.

Overview: Same aggregated topology as §1.1, plus OpenTelemetry tracing instrumentation. Adds `DYN_LOGGING_JSONL=true` and `OTEL_EXPORT_ENABLED=true` at the `DynamoGraphDeployment` level, and per-service `OTEL_SERVICE_NAME` env vars (`dynamo-frontend`, `dynamo-worker-vllm`).

Prerequisites: Chapter 0 complete, plus an OpenTelemetry collector reachable from the cluster (the template's default `OTEL_EXPORTER_OTLP_TRACES_ENDPOINT` points at `http://tempo.observability.svc.cluster.local:4317` — a Grafana Tempo instance in an `observability` namespace; update or remove this if you use a different backend or haven't deployed one). See `docs/observability/tracing.md` in the repo for collector setup.

Deployment Steps:

```
# Edit OTEL_EXPORTER_OTLP_TRACES_ENDPOINT first if your collector endpoint differs
kubectl apply -f examples/backends/vllm/deploy/xpu/agg_tracing_xpu_dra.yaml -n ${NAMESPACE}
```

Verification / Inference Test: same as §1.1, but against Deployment `vllm-agg-tracing-xpu-frontend`. Additionally confirm traces are arriving at your collector (e.g. search for service name `dynamo-worker-vllm` in Tempo/Jaeger UI).

Troubleshooting: - No traces appearing: confirm `OTEL_EXPORTER_OTLP_TRACES_ENDPOINT` is reachable from the pod's network namespace (`kubectl exec` into the pod and `curl/nc` the endpoint). - If you have no observability stack deployed, remove the tracing env vars rather than pointing at a non-existent endpoint, to avoid startup delays from repeated failed export attempts.

Cleanup: `kubectl delete -f ../agg_tracing_xpu_dra.yaml -n ${NAMESPACE}`

Config Reference:

Field	Value	Notes
DynamoGraphDeployment name	vllm-agg-tracing-xpu	
Extra env (deployment-level)	DYN_LOGGING_JSONL=true, OTEL_EXPORT_ENABLED=true, OTEL_EXPORTER_OTLP_TRACES_ENDPOINT	
Extra env (Frontend)	OTEL_SERVICE_NAME=dynamo-frontend	
Extra env (Worker)	OTEL_SERVICE_NAME=dynamo-worker-vllm	

1.3 Aggregated + KV Router (agg_router_xpu_dra.yaml)

Status: documented from repo, not yet run hands-on.

Overview: Adds Dynamo’s KV-aware Router (DYN_ROUTER_MODE=kv on the Frontend) in front of 2 aggregated worker replicas. **Important upstream-documented limitation:** in Aggregated mode, VllmDecodeWorker does not emit BlockStored KV cache events (those are only produced during the prefill phase in Disaggregated mode), so the KV Router **cannot** make cache-aware routing decisions here — it effectively falls back to simpler load-based routing. For genuine KV-aware routing, use §1.8 (disagg_router_xpu_dra.yaml) instead. This case is documented for completeness, but readers who actually need KV-aware routing should go straight to §1.8.

Prerequisites: Chapter 0 complete.

Deployment Steps:

```
kubectl apply -f examples/backends/vllm/deploy/xpu/agg_router_xpu_dra.yaml -n ${NAMESPACE}
```

Verification / Inference Test: same pattern as §1.1, Deployment name xpu-agg-router-frontend, 2 worker replicas.

Troubleshooting: If you expected KV-cache-aware load balancing and don’t see it, this is expected behavior per the limitation above — not a bug.

Cleanup: `kubectl delete -f ../agg_router_xpu_dra.yaml -n ${NAMESPACE}`

Config Reference:

Field	Value	Notes
DynamoGraphDeployment name	xpu-agg-router	
DYN_ROUTER_MODE	kv (Frontend)	limited effectiveness in aggregated mode, see above
Worker replicas	2	
--kv-events-config	ZMQ publisher on tcp://*:20080, topic kv-events	
--enable-prefix-caching	set	

1.4 Aggregated + KV Router, No-KV-Events Mode (agg_router_kv_approx_xpu_dra.yaml)

Status: documented from repo, not yet run hands-on.

Overview: A variant of §1.3 using `--no-kv-events` on the Frontend (`dynamo.frontend --router-mode kv --no-kv-events`) and `enable_kv_cache_events: false` on the worker. Instead of consuming real KV events over NATS/JetStream, the router predicts cache state locally from its own routing decisions with TTL-based expiration/pruning. Benefits per upstream comments: no NATS/JetStream dependency, simpler deployment for lightweight scenarios.

Prerequisites: Chapter 0 complete. No message-bus (NATS) dependency needed for this variant.

Deployment Steps:

```
kubectl apply -f examples/backends/vllm/deploy/xpu/agg_router_kv_approx_xpu_dra.yaml -n ${NAMESPACE}
```

Verification / Inference Test: same pattern as §1.1, Deployment name `xpu-agg-kvap-frontend`.

Troubleshooting: Because routing is based on local prediction rather than real KV events, expect the router’s cache-hit assumptions to drift under high churn — this is an inherent trade-off of the approximate mode, not a misconfiguration.

Cleanup: `kubectl delete -f ../agg_router_kv_approx_xpu_dra.yaml -n ${NAMESPACE}`

Config Reference:

Field	Value	Notes
DynamoGraphDeployment name	<code>xpu-agg-kvap</code>	
Frontend command	<code>dynamo.frontend --router-mode kv --no-kv-events</code>	
<code>--kv-events-config</code>	<code>{"enable_kv_cache_events": false}</code>	

1.5 Disaggregated (disagg_xpu_dra.yaml) — Deep-Dive Sample Case

Status: documented from repo, not yet run hands-on. (Written as the reference sample for this whitepaper; follow this structure when completing the remaining chapters.)

Overview: True Prefill/Decode disaggregation on Intel XPU. Two distinct worker roles (`VllmPrefillWorker`, `VllmDecodeWorker`), each with `subComponentType: prefill/decode`, and KV cache handed off between them over NIXL (`kv_connector: NixlConnector`) with `kv_buffer_device: xpu` (i.e., the KV transfer buffer itself lives in XPU device memory, not host/CPU RAM).

Prerequisites: Chapter 0 complete. Both prefill and decode workers need their own XPU allocation (2 total GPU claims for this template as shipped, since both replicas default to 1).

Deployment Steps:

```
kubectl apply -f examples/backends/vllm/deploy/xpu/
disagg_xpu_dra.yaml -n ${NAMESPACE}
```

Verification:

```
kubectl get resourceclaim -n ${NAMESPACE}           # expect 2 claims
(prefill + decode)
kubectl get dynamographdeployment -n ${NAMESPACE}
kubectl get pods -n ${NAMESPACE}
kubectl logs deployment/vllm-disagg-xpu-dra-vllmprefillworker -n $
${NAMESPACE}
kubectl logs deployment/vllm-disagg-xpu-dra-vllmdecodeworker -n $
${NAMESPACE}
```

Inference Test:

```
kubectl port-forward deployment/vllm-disagg-xpu-dra-frontend
8000:8000 -n ${NAMESPACE}
curl localhost:8000/v1/models
curl localhost:8000/v1/completions \
  -H "Content-Type: application/json" \
  -d '{"model": "Qwen/Qwen3-0.6B", "prompt": "Explain prefill/decode
disaggregation", "max_tokens": 50}'
```

Troubleshooting: - KV transfer failures between prefill and decode workers typically show up as NIXL connection errors in the decode worker's log — confirm both workers were scheduled with XPU devices (`kubectl describe resourceclaim`) and that `kv_buffer_device` matches on both sides (`kv_role: kv_both` on prefill, `kv_role: kv_consumer` on decode — both set to "xpu"). - If either worker Pending forever, check `kubectl get resourceslices` for available device count — you need at least 2 free XPU devices across the cluster for this template.

Cleanup: `kubectl delete -f examples/backends/vllm/deploy/xpu/`
`disagg_xpu_dra.yaml -n ${NAMESPACE}`

Config Reference:

Field	Value	Notes
DynamoGraphDeployment name	vllm-disagg-xpu-dra	
Services	Frontend, VllmPrefillWorker, VllmDecodeWorker (1 replica each)	

Field	Value	Notes
kv_connector	NixlConnector	
Prefill kv_role	kv_both	
Decode kv_role	kv_consumer	
kv_buffer_device	xpu	KV buffer lives in XPU memory, not CPU RAM
--block-size	64	
Total GPU claims	2 (1 prefill + 1 decode)	

1.6 Disaggregated + Tracing (disagg_tracing_xpu_dra.yaml)

Status: documented from repo, not yet run hands-on.

Overview: Same disaggregated topology as §1.5, plus the same OpenTelemetry tracing pattern as §1.2 — DYN_LOGGING_JSONL/OTEL_EXPORT_ENABLED at the deployment level, and per-service OTEL_SERVICE_NAME (dynamo-frontend, dynamo-worker-decode, dynamo-worker-prefill).

Prerequisites: Chapter 0 complete + observability collector as in §1.2.

Deployment Steps:

```
kubectl apply -f examples/backends/vllm/deploy/xpu/
disagg_tracing_xpu_dra.yaml -n ${NAMESPACE}
```

Verification / Inference Test: same pattern as §1.5, Deployment name vllm-disagg-tracing-xpu-frontend.

Troubleshooting: same as §1.2 (tracing endpoint reachability) and §1.5 (KV transfer).

Cleanup: `kubectl delete -f ../disagg_tracing_xpu_dra.yaml -n ${NAMESPACE}`

Config Reference: identical to §1.5 plus the tracing env vars from §1.2, applied per-role (dynamo-worker-decode / dynamo-worker-prefill service names instead of a single dynamo-worker-vllm).

1.7 Disaggregated + Planner

(disagg_planner_xpu_dra.yaml)

Status: documented from repo, not yet run hands-on.

Overview: Adds a Planner component that consumes pre-profiled throughput/latency curves (shipped as a ConfigMap named planner-profile-data, containing prefill_raw_data.json and decode_raw_data.json) to make scaling decisions. The template ships **example** profiling data points (ISL/TTFT/throughput-per-GPU for prefill; KV-usage/context-length/ITL/throughput for decode) — these are illustrative sample curves, not measured on real Intel XPU hardware, and must be replaced with your own profiling results before using the Planner for real capacity decisions (see dynamo profile tooling in the repo, if targeting production use).

Prerequisites: Chapter 0 complete, plus (recommended) your own hardware-profiled prefill_raw_data.json/decode_raw_data.json if you intend to trust the Planner’s scaling recommendations rather than just exercising the deployment mechanics.

Deployment Steps:

```
kubectl apply -f examples/backends/vllm/deploy/xpu/
disagg_planner_xpu_dra.yaml -n ${NAMESPACE}
```

Verification:

```
kubectl get configmap planner-profile-data -n ${NAMESPACE}
kubectl get pods -n ${NAMESPACE}      # expect Frontend, Planner,
VllmDecodeWorker, VllmPrefillWorker
kubectl logs deployment/vllm-disagg-planner-xpu-planner -n $
${NAMESPACE}
```

Inference Test: same as §1.5, Deployment name vllm-disagg-planner-xpu-frontend.

Troubleshooting: - The Planner’s --config JSON hard-codes prefill_engine_num_gpu: 1 and decode_engine_num_gpu: 1 — the upstream comment flags this as a “KEY FIX: Add GPU counts here for DRA fallback” workaround, because the Planner cannot auto-discover GPU counts from DRA ResourceClaims the way it can from resources.limits.gpu on NVIDIA. If you change replica counts or TP size, update these values manually or the Planner’s scaling math will

be wrong. - throughput_adjustment_interval_seconds: 60 controls how often the Planner re-evaluates scaling — reduce for faster reaction in a demo, increase to avoid thrashing in production.

Cleanup: `kubectl delete -f examples/backends/vllm/deploy/xpu/disagg_planner_xpu_dra.yaml -n ${NAMESPACE}`

Config Reference:

Field	Value	Notes
DynamoGraphDeployment name	vllm-disagg-planner-xpu	
Extra component	Planner (dynamo.planner, dynamo-planner:my-tag image)	CPU-only, no XPU claim
prefill_engine_num_gpu / decode_engine_num_gpu	1 / 1	manual DRA fallback values — must match actual TP size
Profile data	ConfigMap/ planner-profile-data	replace sample curves with real Intel XPU profiling data
throughput_adjustment_interval_seconds	60	

1.8 Disaggregated + KV Router

(disagg_router_xpu_dra.yaml)

Status: documented from repo, not yet run hands-on.

Overview: The combination that upstream explicitly recommends for genuine KV-aware routing on Intel XPU — disaggregated P/D (so real BlockStored KV events are emitted during prefill) plus DYN_ROUTER_MODE=kv on the Frontend, --kv-events-config on prefill/decode workers, and --enable-prefix-caching. 2 decode worker replicas by default (vs. 1 for the plain disagg case in §1.5) to give the router something to route between.

Prerequisites: Chapter 0 complete.

Deployment Steps:

```
kubectl apply -f examples/backends/vllm/deploy/xpu/
disagg_router_xpu_dra.yaml -n ${NAMESPACE}
```

Verification:

```
kubectl get pods -n ${NAMESPACE}    # expect Frontend, 2x
VllmDecodeWorker, VllmPrefillWorker
kubectl get resourceclaim -n ${NAMESPACE}    # expect 3 claims
total
```

Inference Test: same as §1.5, Deployment name xpu-disagg-router-frontend. Send several concurrent requests with overlapping prompt prefixes and observe (via logs/metrics) that requests sharing a KV prefix tend to route to the same decode worker.

Troubleshooting: same KV-transfer and DRA-scheduling checks as §1.5; additionally, if routing doesn't appear cache-aware, confirm --kv-events-config is actually enabled on the **prefill** worker (KV events are emitted during prefill, not decode, per the note in §1.3).

Cleanup: kubectl delete -f examples/backends/vllm/deploy/xpu/disagg_router_xpu_dra.yaml -n \${NAMESPACE}

Config Reference:

Field	Value	Notes
DynamoGraphDeployment name	xpu-disagg-router	

Field	Value	Notes
DYN_ROUTER_MODE	kv (Frontend)	fully effective here (unlike §1.3), since prefill emits real KV events
Decode worker replicas	2	
Total GPU claims	3 (2 decode + 1 prefill)	

Chapter 2 — Global Planner on Intel XPU

Status: documented from repo, not yet run hands-on.

Source: `examples/global_planner/global-planner-vllm-test-xpu-dra.yaml`, documented in `examples/global_planner/README.md`.

Overview: A single-endpoint, multi-pool deployment pattern: one Frontend, a GlobalRouter, and a GlobalPlanner, fronting **2 TP1 prefill pools + 1 TP1 decode pool** on Intel XPU (the XPU/DRA variant uses TP1 everywhere because each DRA ResourceClaimTemplate requests exactly one XPU device per worker — there is no multi-GPU-per-claim tensor-parallel config in this template). This differs from Chapter 1’s simpler single-DGD templates by demonstrating Dynamo’s cluster-wide GPU budget / multi-pool planning capability rather than a single fixed topology.

Prerequisites: Chapter 0 complete, plus an **RWX (ReadWriteMany) StorageClass** available in the cluster (`STORAGE_CLASS_NAME` env var below) for shared state between planner components.

Deployment Steps:

```
export K8S_NAMESPACE=my-ns
export DYNAMO_IMAGE=<dynamo-image>
export DYNAMO_VLLM_IMAGE=<vllm-xpu-image>    # from §0.3, must be
built for XPU
export MODEL_NAME=Qwen/Qwen3-0.6B
export STORAGE_CLASS_NAME=<rw-storage-class>
```

```
kubectl create secret generic hf-token-secret \
  --from-literal=HF_TOKEN=<your-token> -n ${K8S_NAMESPACE}

envsubst < examples/global_planner/global-planner-vllm-test-xpu-
dra.yaml \
  | kubectl apply -n ${K8S_NAMESPACE} -f -
```

Verification:

```
kubectl get dynamographdeployment -n ${K8S_NAMESPACE}
kubectl get pods -n ${K8S_NAMESPACE}
kubectl get resourceclaim -n ${K8S_NAMESPACE}
```

Inference Test: port-forward the Frontend service and send completion requests as in §1.5, observing (via GlobalRouter/GlobalPlanner logs) how requests are distributed across the 2 prefill pools and 1 decode pool.

Troubleshooting: - envsubst silently leaves variables unset if you forget to export one of them — always echo the rendered YAML or diff against the template before applying if a field looks empty. - Because every pool is TP1, do not attempt to scale a single pool’s tensor-parallel size without first re-checking whether the DRA ResourceClaimTemplate needs to request more than 1 device — the shipped template does not do this.

Cleanup:

```
envsubst < examples/global_planner/global-planner-vllm-test-xpu-
dra.yaml \
  | kubectl delete -n ${K8S_NAMESPACE} -f -
```

Config Reference:

Field	Value	Notes
Pattern	Single-endpoint, multi-pool	Frontend + GlobalRouter + GlobalPlanner
Prefill pools	2× TP1	XPU/DRA variant, 1 device per pool
Decode pool	1× TP1	
Required extra	RWX StorageClass	for planner shared state
Key difference from GPU (NVIDIA) variant	DRA ResourceClaimTemplate instead of	

Field	Value	Notes
	<code>resources.limits.gpu;</code> <code>VLLM_TARGET_DEVICE=xpu;</code> TP1-only pools	

Chapter 3 — Recipe: Qwen3-VL-32B-FP8 Heterogeneous Hardware Disaggregation (Intel XPU Encode + NVIDIA GPU Decode)

Status: documented from official repo sources, not yet run hands-on.

Overview

`recipes/qwen3-vl-32b-fp8/` is a production-recipe deployment (distinct from the `examples/` tree) for **Qwen/Qwen3-VL-32B-Instruct-FP8**, a 32B FP8-quantized vision-language model. It ships two configurations:

Configuration	Hardware	Mode
<code>vllm/agg/</code>	1× NVIDIA H100/ H200	Aggregated (vision + decode combined) — NVIDIA only, no Intel XPU
<code>vllm/ hetero_hardware_disagg/</code>	1× Intel XPU + 1× NVIDIA GPU	Disaggregated — this is the Intel XPU case

Unlike every other Intel XPU case in this whitepaper (the 8 DRA templates in Chapter 1, which run **entirely** on Intel XPU), this recipe is **heterogeneous**: it deliberately splits the pipeline across two different accelerator vendors in the same deployment —

- **EncodeWorker** — runs the vision-encoding stage on **Intel XPU** (gpu.intel.com DRA device class), producing image embeddings.
- **VllmDecodeWorker** — runs autoregressive text decode on **NVIDIA GPU** (gpu.nvidia.com DRA device class), consuming those embeddings.
- Embeddings are transferred between the two workers over **RDMA** via a NIXL NixlConnector (kv_buffer_device: xpu on the encode side, kv_buffer_device: cuda on the decode side), using UCX_TLS that includes both ze_copy (Intel Level-Zero copy transport) and cuda_copy.

Source: recipes/qwen3-vl-32b-fp8/README.md, vllm/hetero_hardware_disagg/deploy.yaml, vllm/hetero_hardware_disagg/intel_xpu_rdma_template.yaml, vllm/hetero_hardware_disagg/nvidia_gpu_rdma_template.yaml.

Why this differs from Chapter 1’s DRA templates

	Chapter 1 (8 DRA templates)	This recipe
Location	examples/backends/vllm/deploy/xpu/	recipes/qwen3-vl-32b-fp8/vllm/hetero_hardware_disagg/
Hardware	100% Intel XPU	Intel XPU (encode) + NVIDIA GPU (decode) in the same deployment
Purpose	General serving patterns (aggregated/disaggregated × tracing/KV-router/planner) for any model	A specific production recipe for one model (Qwen3-VL-32B-FP8), demonstrating cross-vendor hardware disaggregation
DRA device class	gpu.intel.com only	gpu.intel.com (encode) + gpu.nvidia.com

Chapter 1 (8 DRA templates)	This recipe
	(decode), both requesting rdma-dranet

If your goal is “run Dynamo entirely on Intel XPU,” use Chapter 1. If your goal is “offload multimodal encode to Intel XPU while keeping decode on existing NVIDIA GPU capacity” (e.g. to free up NVIDIA GPU capacity for decode-bound workloads while reusing available Intel XPU nodes for the comparatively lighter encode stage), this recipe is the direct reference.

Prerequisites

- Dynamo Platform installed (see [Common Installation](#)).
- A cluster with **both** an Intel XPU node/pool and an NVIDIA GPU node/pool, each exposed via Kubernetes DRA (`gpu.intel.com` and `gpu.nvidia.com` DeviceClasses respectively).
- **RDMA-capable network interfaces** between the Intel XPU and NVIDIA GPU nodes — this is a hard requirement, not optional; the recipe explicitly depends on a `rdma-dranet` DRA device class on both sides for the NIXL embedding transfer.
- A HuggingFace token with access to Qwen models, stored as `hf-token-secret`.
- A StorageClass supporting `ReadWriteMany` for the shared model cache PVC.

Deployment Steps

```
export NAMESPACE=dynamo-demo
kubectl create namespace ${NAMESPACE}

kubectl create secret generic hf-token-secret \
  --from-literal=HF_TOKEN="<your-token>" \
  -n ${NAMESPACE}

# 1. Update storageClassName in model-cache/model-cache.yaml to
match your cluster, then:
kubectl apply -f model-cache/ -n ${NAMESPACE}
kubectl wait --for=condition=Complete job/model-download -n $
${NAMESPACE} --timeout=3600s

# 2. Apply the DRA ResourceClaimTemplates (both, before the
deployment)
```

```
kubectl apply -f vllm/hetero_hardware_disagg/
intel_xpu_rdma_template.yaml -n ${NAMESPACE}
kubectl apply -f vllm/hetero_hardware_disagg/
nvidia_gpu_rdma_template.yaml -n ${NAMESPACE}
```

3. Deploy

```
kubectl apply -f vllm/hetero_hardware_disagg/deploy.yaml -n $
{NAMESPACE}
```

Verification

```
kubectl get pods -n ${NAMESPACE} -l nvidia.com/dynamo-graph-
deployment-name=qwen3-vl-32b-fp8-vllm-disagg
kubectl get resourceclaims -n ${NAMESPACE}
kubectl logs -n ${NAMESPACE} deploy/qwen3-vl-32b-fp8-vllm-disagg-
encodeworker | grep -i "nixl\|xpu"
kubectl logs -n ${NAMESPACE} deploy/qwen3-vl-32b-fp8-vllm-disagg-
vllmdecodeworker | grep -i "nixl\|cuda"
```

Inference Test

```
kubectl port-forward svc/qwen3-vl-32b-fp8-vllm-disagg-frontend
8000:8000 -n ${NAMESPACE}
```

Text-only request

```
curl http://localhost:8000/v1/chat/completions \
-H "Content-Type: application/json" \
-d '{
  "model": "Qwen/Qwen3-VL-32B-Instruct-FP8",
  "messages": [{"role": "user", "content": "Hello!"}],
  "max_tokens": 50
}'
```

Multimodal (image) request – exercises the Intel XPU encode path

```
curl http://localhost:8000/v1/chat/completions \
-H "Content-Type: application/json" \
-d '{
  "model": "Qwen/Qwen3-VL-32B-Instruct-FP8",
  "messages": [
    {
      "role": "user",
      "content": [
        {"type": "image_url", "image_url": {"url": "https://
upload.wikimedia.org/wikipedia/commons/thumb/4/47/
PNG_transparency_demonstration_1.png/300px-
PNG_transparency_demonstration_1.png"}},
        {"type": "text", "text": "Describe this image in
detail."}
      ]
    }
  ]
}'
```

```

    ]
  }
],
"max_tokens": 256
}'

```

Benchmarking

```
./benchmark/run-benchmark.sh --config disagg -n ${NAMESPACE}
```

Troubleshooting

- **ResourceClaimTemplate not found errors on deploy:** both `intel_xpu_rdma_template.yaml` and `nvidia_gpu_rdma_template.yaml` must be applied *before* `deploy.yaml` — the `deploy.yaml` references both templates by name (`intel-xpu-rdma-template`, `nvidia-gpu-rdma-template`).
- **NIXL handshake failures between encode and decode workers:** confirm RDMA connectivity between the two nodes independently of Dynamo first (e.g. `ib_write_bw/ucx_perftest`), and check that UCX_TLS includes the transport for each side (`ze_copy` for Intel XPU, `cuda_copy` for NVIDIA).
`kv_connector_extra_config.enforce_handshake_compat: false` is already set to relax version matching between the two heterogeneous vLLM builds.
- **EncodeWorker OOM / low throughput:**
`--gpu-memory-utilization 0.7` on the encode side is conservative — adjust based on the Intel XPU device's actual HBM/VRAM size.
- **Model download stuck:** confirm `storageClassName` supports `ReadWriteMany` and the PVC is Bound, not Pending.

Cleanup

```

kubectrl delete -f vllm/hetero_hardware_disagg/deploy.yaml -n ${NAMESPACE}
kubectrl delete -f vllm/hetero_hardware_disagg/
intel_xpu_rdma_template.yaml -n ${NAMESPACE}
kubectrl delete -f vllm/hetero_hardware_disagg/
nvidia_gpu_rdma_template.yaml -n ${NAMESPACE}
kubectrl delete -f model-cache/ -n ${NAMESPACE}
kubectrl delete secret hf-token-secret -n ${NAMESPACE}

```


Config Reference

Field	Value	Notes
Model	Qwen/Qwen3-VL-32B-Instruct-FP8	32B params, FP8-quantized weights and KV cache
Encode image	nvcr.io/nvidia/ai-dynamo/vllm-runtime-xpu:1.2.0	Intel XPU vLLM runtime
Decode image	nvcr.io/nvidia/ai-dynamo/vllm-runtime:1.2.0	standard (NVIDIA) vLLM runtime
Intel XPU DRA device class	gpu.intel.com	requested alongside rdma-dranet in intel-xpu-rdma-template
NVIDIA GPU DRA device class	gpu.nvidia.com	requested alongside rdma-dranet in nvidia-gpu-rdma-template
Embedding transfer	NIXL NixlConnector, DYN_VLLM_EMBEDDING_TRANSFER_MODE=nixl-read	encode kv_buffer_device: xpu, decode kv_buffer_device: cuda
Encode-side UCX_TLS	ib,rc,ud,rc_verbs,ud_verbs,ze_copy	ze_copy = Intel Level-Zero copy transport
Decode-side UCX_TLS	ib,rc,ud,rc_verbs,ud_verbs,cuda_copy	
Decode-side KV cache dtype	fp8	
Shared memory per worker	80Gi	both Encode and Decode workers

Chapter 4 — Feasible but Not Yet Supported on Intel XPU

Answering “what could be supported but isn’t yet?”

Within `examples/backends/vllm/deploy/`, only 8 of the ~20 `DynamoGraphDeployment` templates have an Intel XPU (`xpu/`) counterpart (see Chapter 1). The other vLLM-backend templates below use the same vLLM engine already proven on Intel XPU, with no hardware-specific blocker found — these are gaps, not fundamental limitations:

Template	Path	Why it looks feasible on Intel XPU
LoRA adapter serving	<code>backends/vllm/deploy/lora/agg_lora.yaml</code> , <code>lora/multimodal/agg_qwen_lora.yaml</code>	Plain vLLM, no CUDA-specific field. Already confirmed working on Intel XPU via the bash launch-script path (<code>backends/vllm/launch/lora/xpu/agg_lora_xpu.sh</code>) — the K8s YAML gap is just a templating gap
GAIE (Gateway API Inference Extension) integration	<code>backends/vllm/deploy/gaie/{agg,disagg}.yaml</code> + <code>http-route.yaml</code>	Same routing-layer pattern already proven hardware-agnostic on the llm-side of this whitepaper
Multi-node disaggregation	<code>backends/vllm/deploy/disagg-multinode.yaml</code>	Same NIXL prefill/decode split as the XPU-proven <code>xpu/disagg_xpu_dra.yaml</code> just spread across nodes
GPU Memory Service (GMS)		Kubernetes DRA itself is vendor-

Template	Path	Why it looks feasible on Intel XPU
sidecar / failover	backends/vllm/deploy/agg_gms.yaml, agg_failover.yaml, gms-failover.yaml	neutral (Intel already has its own DRA driver, used by all 8 Chapter 1 templates). The real dependency is GMS own device backend — its design doc (lib/gpu_memory_service/GMS_MULTI_DEVICE.m shows CUDA support done and an Intel XPU backend planned as “Phase 2,” just not implemented yet. Not a hard vendor lock-in, just unfinished
SGLang backend (all patterns)	backends/sglang/deploy/{agg,agg_router,disagg,disagg_planner,disagg-multinode,agg_gms}.yaml	No xpu/ directory anywhere; SGLang engine’s own Intel XPU support not independently verified — flagged as unconfirmed

None of these have been hands-on tested on Intel XPU by this whitepaper’s authors — “feasible” here means “no repo evidence of a hardware blocker,” not “verified working.” Treat as a backlog of candidate work, not a completed case.

Genuinely NVIDIA-specific (real hardware/vendor lock-in found)

These have an explicit, documented dependency that does not currently have an Intel XPU equivalent:

Feature	Path	Lock-in reason
KVBM (multi-tier KV cache: GPU→CPU→SSD→remote)	backends/vllm/ deploy/ agg_kvbm.yaml, disagg_kvbm*.yaml	Intel XPU support is real, active work-in-progress, not yet merged: PRs #7946 and #10520 add SYCL/Level-Zero XPU support to KVBM v2, tracked by DEP issue #9313 , still in design discussion
TensorRT-LLM backend (all patterns, incl. multimodal: Qwen3-VL, Llama4, LLaVA, Qwen2-VL)	backends/trtllm/	TensorRT-LLM is NVIDIA's proprietary compiler/runtime (TensorRT), fundamentally CUDA/NVIDIA-GPU-only — no Intel XPU port exists upstream in TensorRT-LLM itself, so this is a hard vendor lock-in, not a templating gap
Triton Server backend	backends/ tritonserver/	Not investigated in depth for this whitepaper; NVIDIA Triton Inference Server has historically been NVIDIA-GPU-centric, flagged for follow-up rather than asserted

Recipes (recipes/) other than the Chapter 3 hetero case

Confirmed via `grep -r xpu recipes/`: no other recipe under `recipes/` references Intel XPU as of this writing. Each recipe would need to be checked individually against the “feasible” criteria above (does it use vLLM/SGLang with no NVIDIA-only sidecar features) before assuming portability.

Also worth noting: llm-d’s own “feasible but not templated” case

The llm-d side of this whitepaper has one directly analogous case: **guides/rl** (`verl-integration.md`, RLHF/GRPO training routing via the llm-d scheduler). Its guide contains **zero** hardware-specific references (no CUDA, NVIDIA, or XPU mentions at all) — it overrides verl’s request routing to “vLLM/SGLang actors on Ray,” the same routing-layer pattern as [Flow Control](#) and [Multi-Model Routing](#), both of which are already confirmed hardware-agnostic and Intel-XPU-compatible in this whitepaper. This makes `guides/rl` a strong feasible-but-untemplated candidate as well, resolving the previously open `llmd-case-rl-scope-check` scope question with a concrete finding rather than leaving it unconfirmed.

If you want to close these gaps

Building and validating any of the “feasible” candidates above on real Intel XPU hardware, then contributing the resulting `xpu/` template upstream, is the most direct way to expand real Intel XPU coverage — following the structural pattern of Chapter 1’s existing 8 templates (Intel DRA device class `gpu.intel.com`, XCCL/ze_copy transport where NIXL/UCX is involved, custom `vllm-runtime-xpu` image).

Appendix A — Environment Variables Reference

Variable	Where set	Purpose
VLLM_TARGET_DEVICE	worker container	Must be xpu for every Intel XPU worker
HF_TOKEN	hf-token-secret (via envFromSecret)	HuggingFace model download auth
DYN_ROUTER_MODE	Frontend container	kv enables KV-aware routing
DYN_LOGGING_JSONL	deployment-level env	JSON-lines structured logging, used with tracing
OTEL_EXPORT_ENABLED	deployment-level env	Enables OpenTelemetry trace export
OTEL_EXPORTER_OTLP_TRACES_ENDPOINT	deployment-level env	OTLP collector endpoint (update per environment)
OTEL_SERVICE_NAME	per-service container env	Service name shown in trace UI

Appendix B — Known Issues / Workarounds

- **DRA GPU count not auto-discovered by Planner (§1.7):** the Planner cannot read GPU counts directly from ResourceClaims the way it reads `resources.limits.gpu` on NVIDIA. Manually set `prefill_engine_num_gpu/decode_engine_num_gpu` in the Planner's `--config JSON` to match your actual per-worker TP size, and keep them in sync whenever you change replica/TP configuration.

- **Aggregated mode + KV Router limitation** (§1.3): `VllmDecodeWorker` in aggregated mode does not emit `BlockStored` KV events (only produced during prefill in disaggregated mode), so KV Router falls back to non-cache-aware routing. Use §1.8 (`disagg_router_xpu_dra.yaml`) if genuine KV-aware routing matters for your use case.
- **Custom XPU image required for every worker** (§0.3): forgetting to build/push `vllm-runtime-xpu` (or leaving the placeholder tag `my-tag` unedited) is the most likely first-run failure — pods will `ImagePullBackOff` or `CrashLoopBackOff` immediately.
- **DRA API version mismatch**: all 8 templates use `apiVersion: resource.k8s.io/v1`. Clusters running older Kubernetes with only `resource.k8s.io/v1alpha3` or `v1beta1` DRA support will fail to apply these templates as-is; upgrade Kubernetes or check the Intel resource drivers' compatibility matrix.

Dynamo Whitepaper — Case Verification Status Tracker

Status legend: `verified working` / `documented from official sources`, `not yet run hands-on` / `outline only` / `blocked` / `not applicable` (no Intel XPU support exists upstream)

Chapter 0 — Common installation

Item	Status	Owner	Notes
Kubernetes + Dynamo Platform install (CRDs + operator)			Rewritten from docs/ kubernetes/ installation- guide.md, Intel- specific notes (skip GPU Operator, use Intel resource drivers) added
Intel resource drivers for Kubernetes +			External dependency, not part of

Item	Status	Owner	Notes
DeviceClass gpu.intel.com			dynamo repo itself
Custom vllm- runtime-xpu image build			From container/ render.py + container/ README.md
HF token secret			Standard step, same pattern across every template

**Chapter 1 — Intel XPU vLLM backend
templates (examples/backends/vllm/deploy/
xpu/)**

#	Case	File	Status	Owner	Notes
1	Aggregated	agg_xpu_dra.yaml			
2	Aggregated + Tracing	agg_tracing_xpu_dra.yaml			Needs an OTEL collec to fully validate
3	Aggregated + KV Router	agg_router_xpu_dra.yaml			KV-aware routing is limited in aggregated mode (see whitepaper §1.3)
4	Aggregated + KV Router (no-KV- events)	agg_router_kv_approx_xpu_dra.yaml			
5	Disaggregated	disagg_xpu_dra.yaml			First deep- dive / sampl case for this whitepaper
6		disagg_tracing_xpu_dra.yaml			

#	Case	File	Status	Owner	Notes
	Disaggregated + Tracing				Sample profiling data is illustrative only, not measured on real Intel XPU hardware — replace before trusting Planner’s scaling decisions
7	Disaggregated + Planner	disagg_planner_xpu_dra.yaml			Recommendation: use this combo for genuine KV-aware routing on Intel XPU
8	Disaggregated + KV Router	disagg_router_xpu_dra.yaml			

Chapter 2 — Global Planner

Case	File	Status	Owner	Notes
Global Planner (multi-pool, TP1)	examples/global_planner/global-planner-vllm-test-xpu-dra.yaml			Requires RWX StorageClass

Chapter 3 — Recipe: Qwen3-VL-32B-FP8 heterogeneous hardware disaggregation

Case	File	Status	Owner	Notes
Hetero XPU (encode) + NVIDIA GPU (decode) disaggregation	recipes/qwen3-vl-32b-fp8/vllm/hetero_hardware_disagg/			Discovered via a follow-up <code>grep -r xpu recipes/</code>

Case	File	Status	Owner	Notes
				scan; requires RDMA connectivity between an Intel XPU node and an NVIDIA GPU node — hardest prerequisite of any case in this whitepaper

Chapter 4 — Feasible but not yet templated for Intel XPU vs. genuinely NVIDIA-locked

Feature	Path	Status
LoRA adapter serving	backends/vllm/ deploy/lora/	feasible, untemplated — plain vLLM, already proven on XPU via bash launch script
GAIE integration	backends/vllm/ deploy/gaie/	feasible, untemplated — same EPP/InferencePool pattern proven hardware-agnostic on llm-d side
Multi-node disaggregation	backends/vllm/ deploy/disagg-multinode.yaml	feasible, untemplated — same NIXL split as XPU-proven xpu/disagg_xpu_dra.yaml, just multi-node
GMS sidecar / failover	backends/vllm/ deploy/agg_gms.yaml,	feasible, in-progress — GMS’s own design doc plans an Intel XPU backend

Feature	Path	Status
	agg_failover.yaml, gms_failover.yaml	(“Phase 2”), not yet built
SGLang backend (all patterns)	backends/sglang/ deploy/	unconfirmed — no XPU mention in dynamo docs; SGLang engine’s own Intel XPU support not independently verified
KVBM (multi-tier KV cache)	backends/vllm/ deploy/ agg_kvbm.yaml, disagg_kvbm*.yaml	active WIP — unmerged PRs #7946, #10520 add XPU support to KVBM v2, tracked by DEP #9313
TensorRT-LLM backend (incl. multimodal)	backends/trtllm/	NVIDIA-locked — [?] TensorRT is a proprietary NVIDIA-only compiler/runtime
Triton Server backend	backends/ tritonserver/	unconfirmed — not investigated in depth
Recipes other than qwen3-vl-32b-fp8	recipes/	no other XPU content found; each would need individual feasibility review

Recommended next steps

1. Validate Chapter 1 cases on a real Intel XPU cluster, starting with **Disaggregated** (§1.5, already deep-dived) since it is the most representative pattern, then **Disaggregated + KV Router** (§1.8) as the “fully featured” case.
2. Stand up a minimal OTEL collector (e.g. Grafana Tempo) to validate the two tracing variants (§1.2, §1.6) end-to-end rather than just confirming the pods start.
3. Profile real Intel XPU prefill/decode throughput curves and replace the sample data in `disagg_planner_xpu_dra.yaml` (§1.7) before treating Planner recommendations as trustworthy.

4. Validate the Chapter 3 heterogeneous recipe hands-on — this is the highest-complexity case (cross-vendor RDMA + NIXL) and the most likely to surface real-world issues not visible from reading the YAML alone.